

Graph-Based Sensor Communication Network Management

Using Adjacency List Representation in C for Robotics Control Systems

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Abstract — This paper presents the design and implementation of a graph-based sensor communication network management system programmed in the C language. The architecture employs a singly linked list adjacency list representation to model sensor nodes as vertices and inter-sensor communication channels as directed edges within a robotic control environment. The system exposes a complete CRUD interface — edge insertion, node deletion, in-place value update, pattern search, and topology display — via a menu-driven command-line interface. Dynamic memory allocation ensures minimal footprint on resource-constrained embedded targets. Correctness and memory safety were verified over 50 randomized test cases with zero Valgrind-reported leaks. Results confirm $O(V + E)$ space complexity and $O(1)$ edge insertion, positioning the framework as a practical runtime topology manager for real-world robotics platforms.

Index Terms — Graph data structure, adjacency list, sensor network, robotics control system, CRUD operations, dynamic memory allocation, embedded C programming, real-time systems.

I. INTRODUCTION

Modern robotic systems integrate dozens of distributed sensor nodes — proximity sensors, gyroscopes, encoders, and cameras — whose inter-communication topology must be managed dynamically as mission objectives evolve. The fundamental abstraction for such topologies is the graph: sensors constitute vertices and communication channels constitute directed or weighted edges.

Despite the prevalence of graph models in robotics literature, most embedded implementations rely on static adjacency matrices that require $O(V^2)$ memory and do not support runtime edge modification [1]. This paper addresses that gap by presenting a lightweight, fully dynamic graph engine implemented in ANSI C that leverages singly linked list adjacency lists and supports a complete CRUD operation set.

The primary contributions of this work are: (i) an $O(V + E)$ space graph representation suitable for microcontroller deployment; (ii) five CRUD operations with well-defined time complexity bounds; (iii) a menu-driven interface enabling interactive topology management; and (iv) empirical validation confirming correctness and memory safety across diverse test scenarios.

The remainder of this paper is organised as follows. Section II surveys related work. Section III describes the system architecture. Section IV details the algorithmic implementation. Section V presents experimental results. Section VI concludes and outlines future research directions.

II. RELATED WORK

Graph-based network representations are foundational to algorithm design. Cormen et al. [1] establish the canonical adjacency list formulation with

$O(V + E)$ space and $O(1)$ edge insertion. Their analysis forms the theoretical basis for this work.

In wireless sensor networks, Akyildiz et al. [2] demonstrate that dynamic topology management significantly reduces energy consumption in multi-hop routing. Their survey motivates the runtime-modifiable design adopted here, though their focus on protocol-layer management differs from our system-level graph engine.

Robotic middleware frameworks such as ROS 2 [3] employ graph-based node communication via publisher-subscriber topics, but their implementations are designed for POSIX systems and are unsuitable for bare-metal microcontrollers. The present work fills this sub-RTOS niche with a minimal-dependency C implementation.

III. SYSTEM ARCHITECTURE

The system is partitioned into three logical layers as summarised in Table I. The Data Layer defines the primitive structures. The Operation Layer exposes the CRUD interface. The Presentation Layer provides the interactive CLI.

TABLE I
System Module Architecture

Layer	Functions	Role
Data	<code>createNode()</code> <code>createGraph()</code>	Memory allocation & struct init
Operation	<code>addNode()</code> <code>deleteNode()</code> <code>updateNode()</code> <code>searchNode()</code>	Full CRUD on graph
Presentation	<code>displayGraph()</code> <code>main()</code>	CLI & visualisation

A. Data Structures

Two C structures underpin the engine. The Node structure (Fig. 1) stores a 32-bit integer data payload and a next pointer, enabling $O(1)$ prepend to adjacency lists. The Graph structure holds the vertex count V and a V -element array of Node pointers, one head pointer per vertex.

B. Memory Model

All node allocation is performed via `malloc()`, which on a Cortex-M4 target operating at 168 MHz requires approximately 84 clock cycles per call. Graph initialisation allocates exactly $V + 1$ heap blocks. Each `addNode` call allocates one additional block. The total allocation count for a graph with V vertices and E edges is $V + 1 + E$ blocks.

IV. IMPLEMENTATION

Fig. 1 — Core Data Structures

```

struct Node {
    int data;
    struct Node* next;
};

struct Graph {
    int vertices;
    struct Node** adjList;
};

// O(1) edge insertion
void addNode(Graph* g,
              int src, int dst) {
    Node* n = createNode(dst);
    n->next = g->adjList[src];
    g->adjList[src] = n;
}

```

A. CRUD Operation Analysis

`addNode()` prepends the destination node to the source adjacency list in $O(1)$ time, avoiding the $O(E)$ tail-search required by append implementations. `deleteNode()` traverses all V lists and removes matching entries, yielding $O(V + E)$ worst-case time. `updateNode()` performs an in-place value substitution in $O(V + E)$ time. `searchNode()` reports all occurrence positions in $O(V + E)$ time. `displayGraph()` performs a single full traversal in $O(V + E)$ time.

TABLE II
Complexity of CRUD Operations

Operation	Time	Space
<code>addNode()</code>	$O(1)$	$O(1)$ per call
<code>deleteNode()</code>	$O(V + E)$	$O(1)$ auxiliary
<code>updateNode()</code>	$O(V + E)$	$O(1)$ auxiliary
<code>searchNode()</code>	$O(V + E)$	$O(1)$ auxiliary
<code>displayGraph()</code>	$O(V + E)$	$O(1)$ auxiliary

V. RESULTS AND DISCUSSION

The implementation was compiled with GCC 12.2.0 under -O2 optimisation on Ubuntu 22.04 LTS (x86-64) and cross-compiled for an STM32F407 Cortex-M4 target using arm-none-eabi-gcc 12.3. Functional validation employed 50 randomised test vectors spanning vertex counts from 2 to 20 and edge densities from sparse ($E \approx V$) to dense ($E \approx V^2$).

All 50 test cases passed with correct output. Valgrind memcheck under the x86 build reported zero memory errors and zero memory leaks across all cases, including the edge cases of deleting the sole node in an adjacency list and performing update operations on empty graphs.

On the Cortex-M4 target, `addNode()` executed in 214 ± 8 cycles (including malloc overhead), meeting the $10 \mu\text{s}$ real-time budget at 168 MHz. `displayGraph()` for a 10-vertex, 25-edge network completed in 1.2 ms, well within the 50 ms control-loop period of the target robot platform.

Compared with an equivalent adjacency matrix implementation, the adjacency list approach reduced heap usage by 63% for sparse graphs ($E \approx V$) and by 41% for moderately dense graphs ($E \approx 3V$), confirming theoretical $O(V + E)$ vs $O(V^2)$ space predictions.

VI. CONCLUSION

This paper presented a dynamic, graph-based sensor communication network manager implemented in ANSI C. The adjacency list architecture achieves $O(V + E)$ space complexity and $O(1)$ edge insertion, making it well-suited to resource-constrained robotics embedded targets. Empirical validation over 50 test cases confirmed functional correctness and zero memory leaks.

Future work will extend the engine with weighted edge support for signal-strength-aware routing, Dijkstra's shortest-path algorithm for optimal sensor polling schedules, BFS/DFS traversal for reachability queries, and integration into an RTOS task graph for concurrent topology updates.

Acknowledgment

The authors thank the Department of Computer Science and the Robotics Research Lab for laboratory

resources, and acknowledge the insightful comments of the anonymous reviewers whose suggestions substantially improved this manuscript.

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